

Pipeline Document

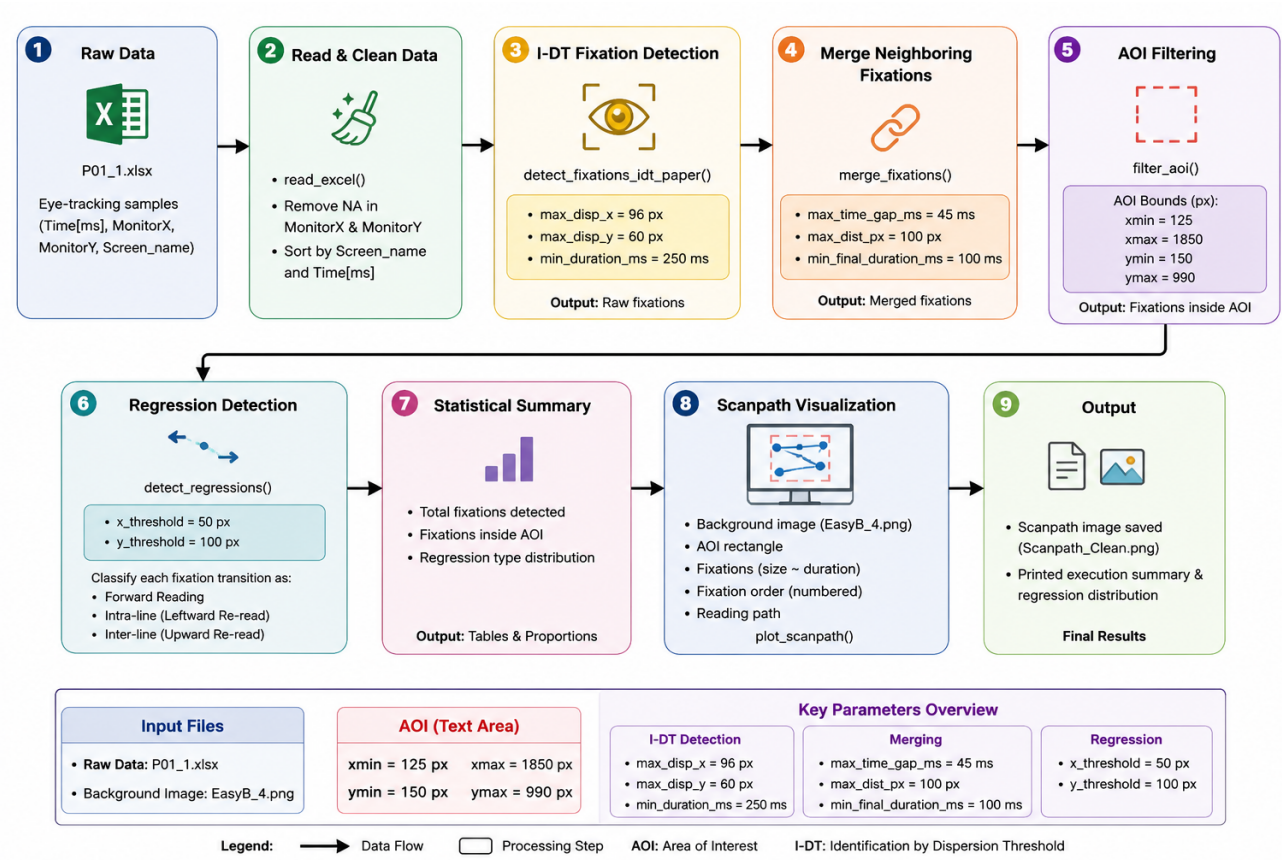
Executive Summary

This script processes raw gaze coordinates (X, Y positions over time) collected from eye-tracking experiments into structured reading eye-movement metrics.

The pipeline performs four key operations:

- Fixation Detection:** Filters raw continuous gaze samples into discrete visual fixations based on spatial dispersion and temporal duration thresholds.
- Fixation Merging:** Combines closely spaced consecutive fixations that represent a single visual pause interrupted by minor noise or blinks.
- AOI Spatial Filtering:** Restricts downstream metrics to fixations occurring within a defined Area of Interest (AOI).
- Regression Classification:** Identifies reading patterns by categorizing movements into progressive forward reading, intra-line re-reading, and inter-line re-reading.

The Flowchart of the Pipeline



Function Explanation

1. detect_fixations_idt_paper(df, max_disp_x, max_disp_y, min_duration_ms)

- Purpose:** Detects primary fixations from raw gaze coordinates using the classic **I-DT algorithm** (Salvucci & Goldberg, 2000).

Parameters

Parameter	Type	Description
df	data.frame	Raw eye-tracking samples. Must contain Screen_name , Time[ms] , MonitorX , and MonitorY columns.
max_disp_x	numeric	Maximum allowed horizontal dispersion (pixels) within a fixation window.
max_disp_y	numeric	Maximum allowed vertical dispersion (pixels) within a fixation window.
min_duration_ms	numeric	Minimum fixation duration in milliseconds required for a valid fixation.

```

Raw Gaze Points          Dispersion Box Test          Detected Fixation
[p1, p2, p3, p4, p5]  ----> Max(X)-Min(X) <= max_disp_x ----> [ Mean_X, Mean_Y ]
                        Max(Y)-Min(Y) <= max_disp_y          Start / End Time

```

- **How it works:**

1. Opens a moving temporal window starting at point i and extending to point j until the duration $(T_j - T_i) \geq \text{min_duration_ms}$.
2. Measures spatial dispersion in X and Y :

$$\text{Disp}_X = \max(X_{i..j}) - \min(X_{i..j})$$

$$\text{Disp}_Y = \max(Y_{i..j}) - \min(Y_{i..j})$$

3. If dispersion is within thresholds ($\text{Disp}_X \leq \text{max_disp_x}$ and $\text{Disp}_Y \leq \text{max_disp_y}$), the window expands point-by-point ($j + 1$) until dispersion breaks threshold.

4. Records the centroid coordinates $(\bar{X}, \bar{Y})$ and advances point i past the fixation ($i = j + 1$).

5. If initial dispersion exceeds threshold, shifts i forward by 1 sample.

- **Key Parameters:**

- `max_disp_x = 96 / max_disp_y = 60` : Bounding pixel limits representing the foveal vision threshold (~1–2 degrees of visual angle).
- `min_duration_ms = 250` : Minimum physiological duration required for visual processing during reading. I selected

About Minimum Duration Selection

- [Lexical complexity and fixation times in reading: Effects of word frequency, verb complexity, and lexical ambiguity](#)
- [Evidence for five types of fixation during a random saccade eye tracking task: Implications for the study of oculomotor fatigue](#)

This article mentions about fixation duration time for high-frequency and low-frequency words.

Condition	First fixation	Gaze duration
High-frequency words	225 ms	243 ms
Low-frequency words	262 ms	330 ms

Fixation duration	Interpretation
< 200 ms	Probably skimming / glance
200–300 ms	Normal reading
300–800 ms	Increased cognitive effort (hard word, sentence integration, possible attentional slowing)
> 800 ms	Candidate prolonged fixation; combine with regressions, blinks, and lack of word-frequency effects before labeling as mind wandering

I've come this conclusion and used 250 ms as minimum duration for the code.

Returns

A `data.frame` containing detected fixations with start time, end time, duration, mean position, and sample count.

2. `merge_fixations(fix_df, max_time_gap_ms, max_dist_px, min_final_duration_ms)`

Parameters

Parameter	Type	Description
<code>fix_df</code>	<code>data.frame</code>	Fixation data to merge.
<code>max_time_gap_ms</code>	numeric	Maximum time gap (milliseconds) allowed between two fixations for merging.
<code>max_dist_px</code>	numeric	Maximum Euclidean distance (pixels) allowed between two fixations for merging.
<code>min_final_duration_ms</code>	numeric	Minimum duration retained after merging; shorter fixations are removed.

- **Purpose:** Merges micro-saccades, minor jitter, or brief blinks that artificially split a single fixation into multiple close fragments.
- **How it works:** Calculates temporal gap and spatial distance between consecutive fixations (F_k and F_{k+1}):

$$\Delta T = T_{\text{start},k+1} - T_{\text{end},k}$$

$$\text{Dist} = \sqrt{(X_{k+1} - X_k)^2 + (Y_{k+1} - Y_k)^2}$$

If $\Delta T \leq \text{max_time_gap_ms}$ and $\text{Dist} \leq \text{max_dist_px}$, fixations are combined. The merged centroid

uses a **duration-weighted average** so longer fixations pull the center of gravity proportionately:

$$\bar{X}_{\text{merged}} = \frac{(X_1 \cdot D_1) + (X_2 \cdot D_2)}{D_1 + D_2}$$

- **Key Parameters:**

- `max_time_gap_ms = 45` : Max duration of a acceptable inter-fixation pause.
- `max_dist_px = 100` : Spatial proximity limit (px) for merging.
- `min_final_duration_ms = 100` : Drops leftover short artifacts post-merge.

Returns

A cleaned `data.frame` of merged fixations.

3. `filter_aoi(fix_df, xmin, xmax, ymin, ymax)`

Parameters

Parameter	Type	Description
<code>fix_df</code>	<code>data.frame</code>	Fixation data generated by the fixation detection stage.
<code>xmin</code>	numeric	Left boundary of the AOI (pixels).
<code>xmax</code>	numeric	Right boundary of the AOI (pixels).
<code>ymin</code>	numeric	Top boundary of the AOI (pixels).
<code>ymax</code>	numeric	Bottom boundary of the AOI (pixels).

- **Purpose:** Bounds analysis strictly to the active reading text box, discarding off-screen gazes or off-text margins.
- **Filtering Condition:**

$$\text{Keep} \iff (x_{\min} \leq \bar{X} \leq x_{\max}) \wedge (y_{\min} \leq \bar{Y} \leq y_{\max})$$

Returns

A filtered `data.frame` containing only fixations located inside the AOI.

4. `detect_regressions(fix_df, x_threshold, y_threshold)`

Parameters

Parameter	Type	Description
<code>fix_df</code>	<code>data.frame</code>	Ordered fixation data containing at least <code>Mean_X</code> , <code>Mean_Y</code> , and <code>Start_Time</code> .
<code>x_threshold</code>	numeric	Minimum leftward movement (pixels) required to classify an intra-line regression.
<code>y_threshold</code>	numeric	Minimum upward movement (pixels) required to classify an inter-line regression.

- **Purpose:** Categorizes eye movements to evaluate reading fluency and text difficulty.
- **Classification Rules:**
 - **Intra-line Regression (Leftward Re-read):** Re-reading words on the same line.

$$\Delta X < -x_threshold \quad \text{AND} \quad |\Delta Y| < y_threshold$$

- **Inter-line Regression (Upward Re-read):** Jumping backward to a previous line above.

$$\Delta Y < -y_threshold$$

- **Forward Reading:** Progressive left-to-right or downward movement.

Returns

A `data.frame` with additional columns describing fixation-to-fixation movement and regression type.

5. `plot_scanpath(fix_df, target_screen, bg_image_path, aoi_bounds)`

Parameters

Parameter	Type	Description
<code>fix_df</code>	<code>data.frame</code>	Fixation data to visualize.
<code>target_screen</code>	character/numeric	Screen identifier to plot.
<code>bg_image_path</code>	character/NULL	Optional path to a background image displayed behind the scanpath.
<code>aoi_bounds</code>	list/NULL	Optional AOI boundaries with <code>xmin</code> , <code>xmax</code> , <code>ymin</code> , and <code>ymax</code> .

- **Purpose:** Visualizes the gaze path over the original stimulus image for qualitative validation.
- **Key Features:**
 - Overlays fixation path nodes (`Fix_ID`) connected sequentially by directional saccade vectors.
 - Adjusts node circle size proportional to fixation duration (`Duration_ms`).
 - Inverts the *Y*-axis using `scale_y_reverse()` to align display coordinates (top-left (0,0) origin) with R's default plot coordinates (bottom-left origin).

Return

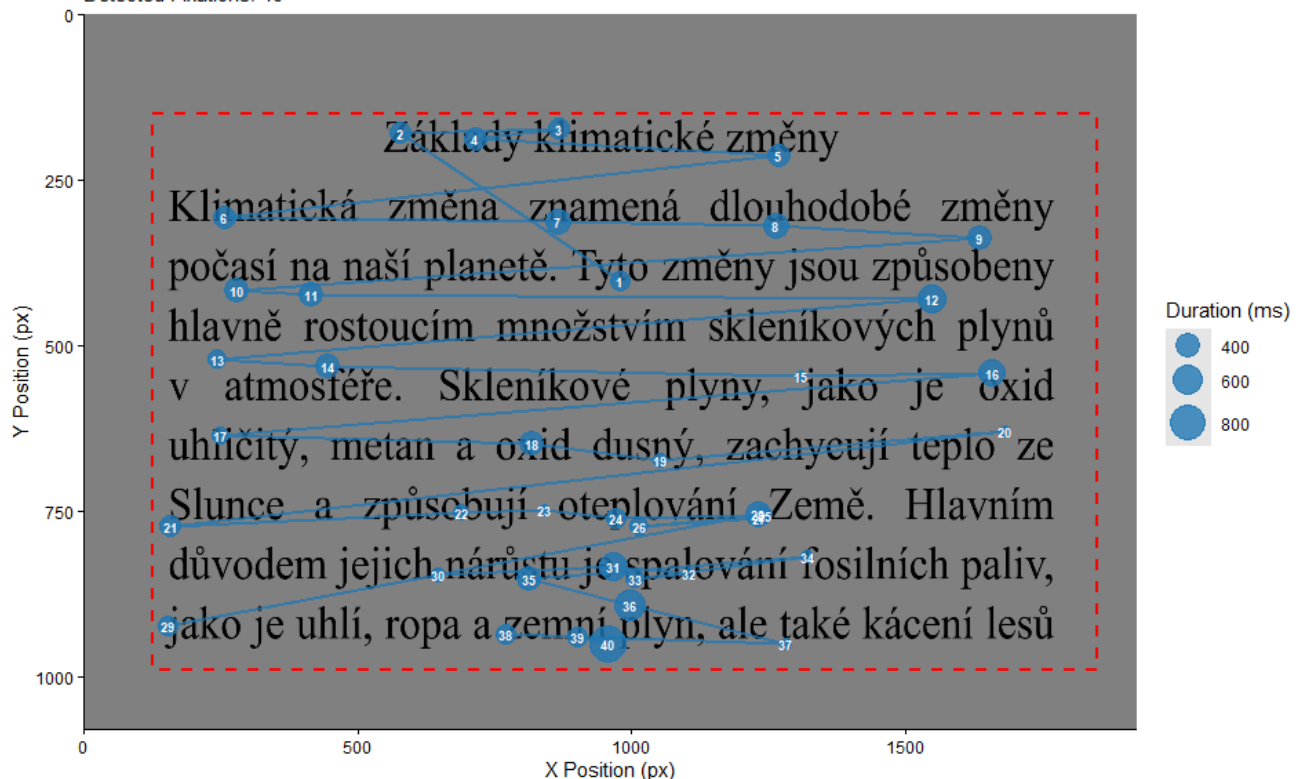
A `ggplot` object representing the scanpath.

Final Result

Fixation scanpath analysis for screen 1:

Scanpath Analysis — Screen 1

Detected Fixations: 40



Fixation counts for each easy screen:

```
> table(fixations_raw$screen_name)

 1  2  3  4
43 45 38 27
> fixations_merged <- fixations_raw %>%
+   merge_fixations(
+     max_time_gap_ms      = 45,
+     max_dist_px          = 100,
+     min_final_duration_ms = 100
+   )
> table(fixations_merged$screen_name)

 1  2  3  4
41 42 37 26
> fixations_final <- fixations_merged %>%
+   filter_aoi(xmin = aoi_box$xmin, xmax = aoi_box$xmax, ymin = aoi_box$ymin, ymax = aoi_box$ymax) %>%
+   detect_regressions(x_threshold = 50, y_threshold = 100)
> cat("\n--- Execution Summary ---\n")

--- Execution Summary ---
> cat("Total Fixations Detected:", nrow(fixations_raw), "\n")
Total Fixations Detected: 153
> cat("Fixations inside Text AOI:", nrow(fixations_final), "\n\n")
Fixations inside Text AOI: 145

> print(table(fixations_final$screen_name, dnn = list("Screen Name")))
Screen Name
 1  2  3  4
40 42 37 26
```